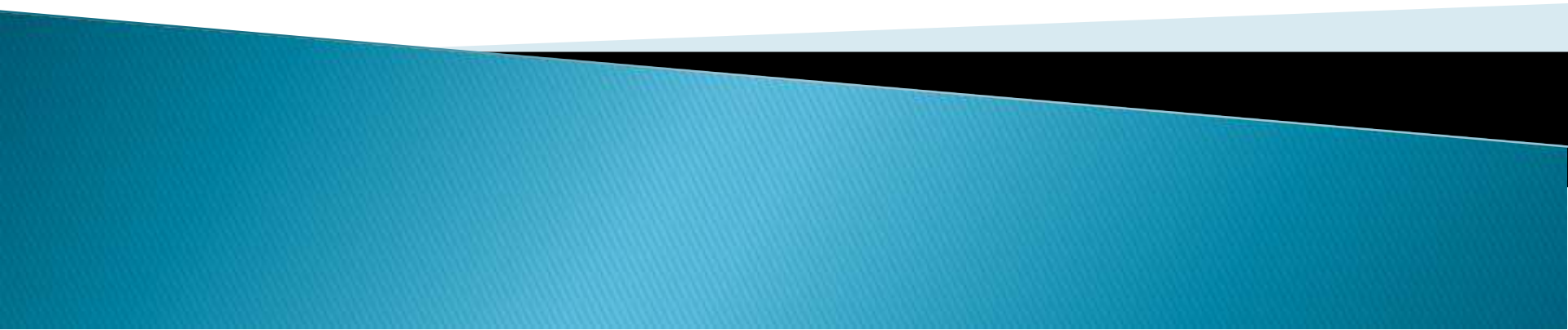


CHAPTER 5:

ENERGY

PREREQUISITE FOR WORK



Introduction

- ▶ Allah Ta'ala states
- ▶ Power belongs wholly to Allah Ta'ala
- ▶ Allah Ta'ala has is omnipotent. He provides energy to every body and every thing.
- ▶ Energy is a meta-physical entity.
- ▶ It does not occupy space nor does it have mass, still it exist in many forms

Cont'd

- ▶ Energy can be defined as ability to do work
- ▶ Energy can not be created nor destroyed but can be transformed from one form to another
- ▶ It is only Allah Ta'ala who can create and has the power to destroy it
- ▶ Note
- ▶ $E = MC^2$
- ▶ Where **E** is energy, **M** is mass and **C** is the speed of light (3.0×10^8)

Energy resources

- ▶ Sun
- ▶ Fossil fuels
- ▶ Nuclear
- ▶ Hydro power (water)
- ▶ Wind
- ▶ Geothermal (steam)
- ▶ Biomass from wood fuel

Forms of energy

- ▶ **Chemical energy**– is the energy stored within a substance through the bonds of chemical compounds. E.g energy stored by fuels such as coal or petrol and is given in form of heat and light
- ▶ **Electrical energy**– energy possessed by moving electrons in a conductor
- ▶ **Magnetic energy**– energy due to repulsion and attraction in magnets
- ▶ **Nuclear energy**–energy due to radioactive elements e.g uranium

Cont'd

- ▶ **Potential energy**– is stored energy. Is divided into two;
 1. ***Gravitational p. e*** – is associated with the energy stored by an object because of its location above the ground (g.p.e)
 2. ***Elastic potential energy*** is the energy stored by any object that can stretch or compress

Cont'd

- ▶ **Strain energy**– energy stored in an elastic object
- ▶ **Internal energy**– energy stored in hot objects
- ▶ **Thermal energy**– energy emitted by hot objects
- ▶ **Light energy**– energy produced by glowing objects
- ▶ **Sound energy**– energy produce to vibrating objects
- ▶ **Kinetic energy (k.e)**– energy possessed by bodies in motion

Transformation and conversion

- ▶ **Energy transformation**, also termed as **energy conversion**, is the process of changing **energy** from one of its forms into another
- ▶ The **conservation of energy principle** states that energy can neither be destroyed nor created. Instead, energy just transforms from one form into another.

Cont'd

▶ Energy can be transferred into four ways

1. By force
2. By heating
3. By radiation
4. By electrical

For instance

Lets take a rocket lifting from the ground

Chemical energy–k.e–g.p.e–thermal energy–light energy–sound energy

Energy efficiency

- ▶ In any energy conversion, the total amount of energy before and after conversion is constant
- ▶ Energy is valuable and we should not waste it
- ▶ Using more energy than necessary increases the damage we do in the environment
- ▶ Most wasted energy ends up thermal energy
- ▶ When fuels are burned they produce heat hence losing energy to the surrounding

Cont'd

- Friction is problem to the moving bodies
- ▶ It can be reduced by lubrication
- ▶ Noisy machinery, loud car engines do waste energy

Making better use of energy

Also referred to as energy conservation

- ▶ Share car/ carpool
- ▶ Reduce waste
- ▶ Plant a tree
- ▶ Do not leave appliances on standby
- ▶ Replace a regular incandescent light bulb with compact fluorescent light bulb
- ▶ Cover your food while cooking
- ▶ Use washing machine or dishwasher only when they are full

Energy calculations

- ▶ Energy is a quantity that can be calculated
- ▶ Kinetic energy is the energy possessed by moving objects

Kinetic energy depends on two factors i.e **mass** and **speed** of the moving body

- ▶ The more the mass, the greater the k.e
- ▶ The more the speed, the greater the k.e

$$K.E = 1/2ms^2$$

Kinetic energy is usually measured in units of **Joules (J)**; one **Joule** is equal to **1 kg m² / s²**

Example

1. A van of mass 200kg is travelling at 10m/s. calculate its kinetic energy. If its speed increases to 20m/s, by how much does kinetic energy increase
2. A car is traveling at a velocity of 10 m/s and is has a mass of 250 Kg. Compute its Kinetic energy?
3. A man is transporting a trolley of mass 6 Kg and having Kinetic energy of 40 J. Compute its Velocity with which he is running?